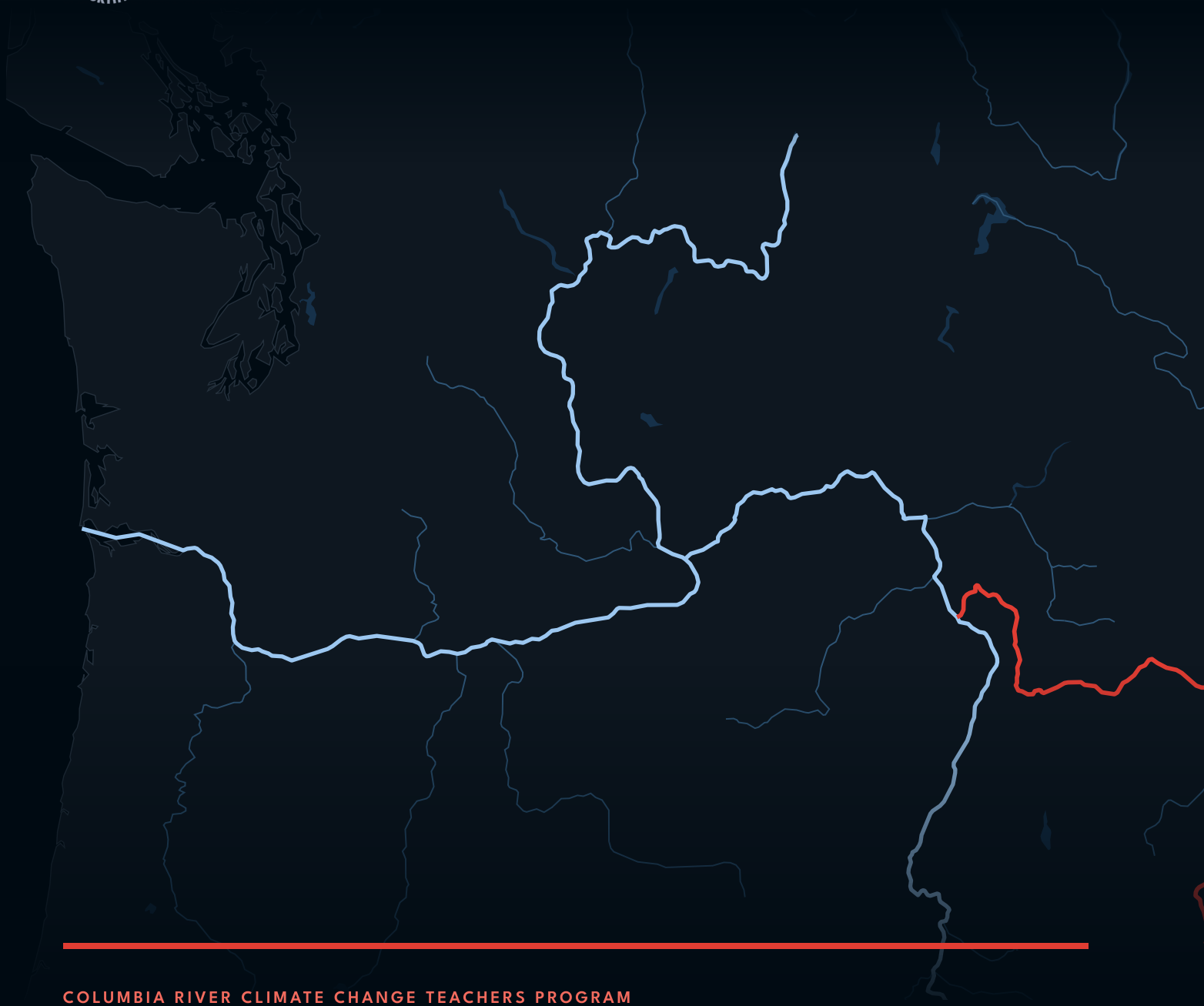
ATNI CLIMATE



COLUMBIA RIVER CLIMATE CHANGE TEACHERS PROGRAM

UNDERSTANDING A CLIMATE OF CHANGE

GUIDING DOCUMENT AND LANGUAGE REVISION

Patrick Freeland and Cayla Jones · ATNI Climate, Affiliated Tribes of Northwest Indians

Workshop printing, 10/20/2026 · Draft for review

FOUR QUESTIONS, ONE RIVER

1	2	3	4
How does a cloud work?	How do you know what you know?	What can we do?	How do we work together?

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Patrick Freeland and Cayla Jones · ATNI Climate, Affiliated Tribes of Northwest Indians

Understanding an Ever Changing Climate: Guiding Document and Language Revision.

Printed 10/20/2026 from the Markdown source in the public repository https://github.com/atniclimate/climate-curriculum. Live workshop: https://atniclimate.github.io/climate-curriculum/

How to Use This Document

This is the guiding document for the series and everything built on it: the 30-minute teacher workshop, the Grades 6 to 12 lessons, CCIC 200, and TCR 201. For each part it holds three layers:

1. **Source text, verbatim.** Copied word for word from the PF Presentation tab of *Climate Change & The Tribes Curriculum 062326*. Where the source has two versions of a passage (an original and a reworded second pass), both appear, labeled "first version" and "second version." Nothing in these blocks has been edited.
2. **Polished language.** A proposed merged version in the series voice, ready for slides, handouts, and webinars. It keeps the source's words wherever they already work and changes only what needed it.
3. **Notes.** Why a change was made, and choices only the authors can make.

The polished language is a proposal. The verbatim text is the authority.

The Series

First version (verbatim):

> Understanding a Climate of Change is a four part series, each about an hour, built so it can stand on its own or move with you at your own pace; also could be made into webinars. It is meant to be understandable by a range of participants, self-guided, but with ways to learn more. The lesson begins with (1) experiential knowledge, and the nature of nature; then (2) explores how we come to know what we know, how knowledges shape our thinking, and what knowledges we should trust more than others, then (3) the differences between adaptation, mitigation, and resilience, and how these actions are better accomplished; and concludes with (4) relationships matter, that it takes all of us to make a difference, and that all of us already do, and that no one can carry all of this work alone.

Sharing, first version (verbatim):

> These lessons are made to be shared broadly, with leaders, teachers, communities, with respectful communication so that both an Elder and a child can understand; the surest sign you understand something is being able to pass it on.There is no test here, only reflections and encouraging inquiry, understanding education is not a race, and that questions leading to questions empower participants to choose what to carry forward.

Sharing, second version (verbatim):

> These lessons are made to be shared with leaders, teachers, and community members. Through respectful communication that both an elder and a child can fully understand; the surest sign that you understand is to teach. There is no test in this curriculum, only reflections and encouragement to understand that education is not a race to debate the rights or wrongs. We must understand that questions leading to empowerment will encourage participants of the importance of knowledge and teachings.

Polished:

Understanding a Climate of Change is a four-part series. Each part runs about an hour and can stand on its own, move at your own pace, or become a webinar; the 30-minute teacher workshop is its short form. It is meant to be understood by a range of participants, self-guided, with ways to learn more. It begins with **(1)** experiential knowledge and the nature of nature; then **(2)** explores how we come to know what we know, how knowledges shape our thinking, and which knowledges we should trust; then **(3)** asks what we can do, through the different work of adaptation, mitigation, and resilience; and closes with **(4)** how we work together: relationships matter, it takes all of us to make a difference, all of us already do, and no one can carry this work alone.

These lessons are made to be shared broadly, with leaders, teachers, and communities, through respectful communication so that both an Elder and a child can understand; the surest sign you understand something is being able to pass it on. There is no test here, only reflection and encouraging inquiry. Education is not a race, and questions that lead to more questions let each of us choose what to carry forward.

Notes: The first version carries the meaning; the second adds "the surest sign that you understand is to teach," which the polished text keeps as "being able to pass it on." The second version's "a race to debate the rights or wrongs" narrows the idea; the polished text keeps the broader "education is not a race." "Elder" is capitalized throughout.

Guiding Line

Verbatim:

> “Think… when both an Elder and a Child can understand, then this is where learning happens.”

Polished: keep as written. It opens the workshop's Four Questions slide.

The Four Questions

Verbatim:

> 1. How does a cloud work?
>
> Have you ever looked up and thought about how (water moves into the sky and comes down as rain and snow?)
>
> 2. How do you know what you know?
>
> Your own senses you can see the snow in the winter, but how do we learn about weather and climate in the classroom and in the media? Senses are in the brain, connect to the graphs, maps, and numbers that scientists and decision-makers use to understand our planet… the moon’s eye view… connect what you can see (in your own backyard) with the big-picture (data?) that help us understand our changing world
>
> 3. What can we do?
>
> When climate change is an issue, and people come to understanding, people take action, we can respond, break down big words like Mitigation, Adaptation, and Resilience into real-world steps you, your family, and your community can take
>
> 4. How do we work together?
>
> Relationships Matter… no single person, or scientific report, or action plan has all the answers. Ways of knowing, Indigenous Knowledge and remote session

Polished:

1. **How does a cloud work?** Have you ever looked up and thought about how water moves into the sky and comes back down as rain and snow?
2. **How do you know what you know?** With your own senses you can see the snow in the winter. In the classroom and in the media, we also learn through graphs, maps, and numbers. This part connects what you can see in your own backyard with the big picture that scientists and decision-makers use to understand our changing world, from the frog's eye to the moon's eye.
3. **What can we do?** When people come to understanding, people take action. We break down big words like mitigation, adaptation, and resilience into real-world steps you, your family, and your community can take.
4. **How do we work together?** Relationships matter. No single person, scientific report, or action plan has all the answers. Many ways of knowing meet here, Indigenous Knowledge among them, in the room and in remote sessions.

Notes: Parts One and Two keep their original titles. Parts Three and Four now lead with a question ("What can we do?" and "How do we work together?"), and the original titles become their subtitles: Adaptation, Mitigation, Resilience; and Relationships Matter. The source's Part One heading, "What is a cloud?", becomes "How does a cloud work?" to match the Four Questions; "What is the function of a cloud?" stays as the subquestion.

Themes

Verbatim:

> Themes: Cycles, relationships, local observation, critical thinking, intergenerational knowledge, change; as more than physical but emotional, scale (moon’s eye view, bird’s eye view, frog’s eye view).

Polished:

- **Cycles:** water, seasons, salmon, and generations all come back around.
- **Relationships:** with each other, and with the lands, waters, and all Beings who live within them.
- **Local observation:** what you can see, hear, and feel where you live.
- **Critical thinking:** thinking about thinking; reading our sources and ourselves.
- **Intergenerational knowledge:** learning from Elders, teaching the young, and the responsibility to do both.
- **Change, felt as well as measured:** change is physical, and it is emotional too.
- **Scale:** the Frog's Eye (up close, at the water), the Bird's Eye (above the land, the whole watershed), and the Moon's Eye (the whole planet, seen through satellites and global records).

1

PART ONE

How Does a Cloud Work?

Source headings, verbatim: "Part One: What is a cloud?" and "What is the function of a cloud?"

KEY LESSON

First version (verbatim):

> Weather is based on fundamental physics, here and throughout the universe. Pressure and water are always moving, never fixed or stuck at one height.

Polished:

Weather is based on fundamental physics, here and throughout the universe. Pressure and water are always moving, never fixed or stuck at one height.

KEY TAKEAWAY

First version (verbatim):

> The same physics and data points that predict tomorrow's weather, also demonstrate patterns and change over a variety of years. With enough data we can look ahead of what's to come, not on speculation, but on factual evidence and supporting science.

Polished:

The same physics and data points that predict tomorrow's weather also show patterns and change over many years. With enough data, we look ahead on evidence and momentum, not speculation.

DESCRIPTION

First version (verbatim):

> We start with a cloud because everyone knows what it is; but many may not have thought through how it “works.” Before we can talk about climate, we have to understand our atmosphere, and ask what it means before we ask how it works: pressure, water, and heat moving through a system that never sits still, dripping like an air conditioner for rain and picking up speed for storms. We will use earth.nullschool to watch it move, because once you see weather as physics you can observe and measure, you can see how the same points of data tell us a story of change over time.

Second version (verbatim):

> We know what a cloud is and what it looks like, but oftentimes we overlook the functions of a cloud, and how it benefits our daily lives. Before we discuss climate, we must first understand our atmospheric functions. Pressure, water, and heat, all fluctuate through a system that never sits idle. We will use earth.nullschool to observe the atmosphere in realtime. Once you observe weather and physics as one, you will be able to measure points of data and their changes over time.

Polished:

We start with a cloud because everyone knows what it is, but many of us have not thought through how it works, or what it does for us every day. Before we can talk about climate, we have to understand our atmosphere: pressure, water, and heat moving through a system that never sits still, dripping like an air conditioner for rain and picking up speed for storms. We use earth.nullschool to watch it move in real time, because once you see weather as physics you can observe and measure, the same points of data tell a story of change over time.

INQUIRY

First version (verbatim):

> What did you learn more deeply about the weather? What may have changed about how you think about clouds, the atmosphere, and/or the water cycle that shifted for you today? What is one question about the weather you want to find the answer to?

Second version (verbatim):

> What have you learned about the weather that deepens your understanding of the atmosphere, the clouds, and the water cycle? What changes shifted in how you think about the weather, the clouds, the atmosphere, and/or the water cycle? Is there anything about the weather that you find yourself wanting to learn more about?

Polished:

What did you learn more deeply about the weather? What shifted in how you think about clouds, the atmosphere, or the water cycle? What is one question about the weather you want to find the answer to?

Notes: The key lesson is kept verbatim. The takeaway trims "factual evidence and supporting science" to "evidence," and keeps "momentum, not speculation," the workshop's phrasing. The description folds in the second version's point that clouds benefit our daily lives.

Guiding questions in the workshop:

- Have you ever looked up and thought about how water moves into the sky and comes down as rain and snow? (Part One title slide)
- Which side of the mountains do you live on, and what does that mean for your rain and snow? (the cloud and landscape model)
- What would you measure where you live, and for how long? (the gauges)

2

PART TWO

How Do You Know What You Know?

Source subheading, verbatim: "Do we really know what we know?"

KEY LESSON

First version (verbatim):

> Climate is social as much as it is scientific; society moves action much more than science. How we come to know things, and who we trust, shapes what we believe.

Second version (verbatim):

> The conversation around climate change is led through social and societal perspectives. Climate is as much social as it is scientific. The actions and conversations in climate are driven behind the knowledge and the systems we trust, they shape what we believe individually and socially.

Polished:

Climate is as much social as it is scientific; society moves action more than science does. How we come to know things, and who we trust, shapes what we believe.

KEY TAKEAWAY

First version (verbatim):

> Narrative and emotion move us whether we notice it or not, so we learn to read both our sources and ourselves. Think about thinking.

Polished:

Narrative and emotion move us whether we notice it or not, so we learn to read both our sources and ourselves. Think about thinking.

DESCRIPTION

First version (verbatim):

> Just like the atmosphere, most of us only partly understand climate, and we fill the gaps with what we are told. Some sources are honest, some carry an agenda, and some push a story they believe is for our own good. This hour is about noticing how narrative and emotion move us, so we can think clearly instead of being moved without knowing it. We have to understand the intergenerational ways-of-knowing, our plural knowledges, our ability to not only learn but to teach. The true mark of understanding something is if you can explain to an Elder and a child so that both understand as well.

Second version (verbatim):

> As we discussed the atmosphere and the climate, we learned that most of us only partly understand how we absorb knowledge and the true viability of the sources we trust. Some sources are honest, some carry an agenda, and some tell a story they believe is for our own good. This hour is focused on noticing narrative and emotion, and how they motivate our way of thinking and teaching. We have to understand intergenerational knowledge and the importance of learning it, but teaching it as well. The true mark of understanding is in how you teach it to our elders and to our children.

Polished:

Just like the atmosphere, most of us only partly understand climate, and we fill the gaps with what we are told. Some sources are honest, some carry an agenda, and some tell a story they believe is for our own good. This part is about noticing how narrative and emotion move us, so we can think clearly rather than be moved without knowing it. We honor intergenerational ways of knowing, our plural knowledges, and our responsibility not only to learn but to teach. The true mark of understanding is explaining something so that both an Elder and a child understand.

INQUIRY

First version (verbatim):

> What did you learn about how we come to know things, how we know these things, and who we trust to share their knowledge with us? Was there an idea you held that has either changed or becomes more clear? What is one question about "climate change" that you want to follow up on?

Second version (verbatim):

> What values did you identify in how we obtain, attain, and carry our knowledge? Was there an idea or a teaching you held that either changed or became more clear after this discussion? What is one question about climate change that you need a better understanding of?

Polished:

What did you learn about how we come to know things, and who we trust to share their knowledge with us? What values guide how we gather and carry knowledge? Was there an idea or a teaching you held that changed, or became more clear? What is one question about climate change you want to follow up on?

Notes: The second version's question about values ("What values did you identify in how we obtain, attain, and carry our knowledge?") is a strong addition and is folded into the polished inquiry. The second version's description ends "how you teach it to our elders and to our children"; the polished text keeps the first version's Elder-and-child test, which the workshop uses as its guiding line.

Guiding questions in the workshop:

- With your own senses you can see the snow in the winter. How do we learn about weather and climate in the classroom, and in the media? (Part Two title slide)
- What can only be seen up close, and what can only be seen from far away? (Frog's Eye, Bird's Eye, Moon's Eye)
- Which one did you feel before you thought about it? (source check)
- What would you need to make this journey, and what would make it harder? (the salmon's journey home)
- Which of these relatives have you seen, eaten, or heard stories about? (relatives of the river)

3

PART THREE

What Can We Do?

Adaptation, Mitigation, Resilience

KEY LESSON

First version (verbatim):

> These three are not the same. They are different jobs with different tools: policy, groundwork, preparation, data, knowledge, funding, and sometimes emergency.

Second version (verbatim):

> Adaptation, mitigation, and resilience are not the same thing. Each has a different job with a different tool set: policy, groundwork, preparation, data knowledge, funding, all are different.

Polished:

These three are not the same. They are different jobs with different tools: policy, groundwork, preparation, data, knowledge, funding, and sometimes emergency.

KEY TAKEAWAY

First version (verbatim):

> The impacts are already here, so we respond quickly and on purpose, without getting lost in fear, the future, and the “black mirror” of media.

Second version (verbatim):

> Climate impacts are already presenting themselves through many forms. We must respond with wit and intent on preserving our knowledge and our teachings without getting too lost in fear of the future, or the “smoke and mirrors” of the media.

Polished:

The impacts are already here, so we respond quickly and on purpose, without getting lost in fear of the future or the "black mirror" of media.

DESCRIPTION

First version (verbatim):

> Adaptation, mitigation, and resilience get used like they are one word, but they are different work done in different ways; thought of in different ways. Climate impacts are happening now, so we must adapt now if we are to be resilient, we must mitigate now if we are to limit the impacts of future generations. We hold onto our sacred traditions, but our livelihoods, and the paradigms of our livelihoods, have to shift and the response has to move faster than it used to. We have to understand data, we have to know what data is useful, we have to know how data leads to action, but we also need to practice good knowledge sharing, respectful communication, so we do not get lost in it, and most importantly we have to be mindful of how we act, and eliminate tedium where it doesn’t matter.

Second version (verbatim):

> Adaptation, mitigation, and resilience are used in terms of likeness and similarity, but each method is different. We must adapt to present climate impacts to mitigate the impacts on our future generations. We must remain resilient in our sacred teachings and traditions, but also in the paradigm of our livelihoods. We have to understand data

> and its usefulness, and how data can lead to action. In any form, we must understand the respectful practice of knowledge sharing.

Polished:

Adaptation, mitigation, and resilience get used like they are one word, but they are different work, done in different ways and thought of in different ways. Climate impacts are happening now, so we must adapt now if we are to be resilient, and we must mitigate now if we are to limit the impacts on future generations. We hold onto our sacred traditions and teachings, while our livelihoods, and the paradigms of our livelihoods, have to shift, and the response has to move faster than it used to. We have to understand data: which data is useful, and how data leads to action. We also practice good knowledge sharing and respectful communication, so we do not get lost in it; and most importantly, we stay mindful of how we act, and cut tedium where it does not matter.

INQUIRY

First version (verbatim):

> How do you now understand climate change adaptation, mitigation, and resilience? Did your sense of what needs to happen, or how fast, change today? What is one question about responding to climate impacts in your own community that you want to find out?

Second version (verbatim):

> Has your understanding of climate adaptation, mitigation, and resilience changed? Has your need for change heightened or increased after our discussion? Are there any questions in regard to climate impacts in your own community and how to respond to them?

Polished:

How do you now understand climate change adaptation, mitigation, and resilience? Did your sense of what needs to happen, or how fast, change today? What is one question about responding to climate impacts in your own community that you want to find out?

Notes: One meaning change to decide. The second version's description says "We must adapt to present climate impacts to mitigate the impacts on our future generations." That joins adaptation and mitigation into one action, which is the confusion Part Three exists to untangle; the polished text keeps the first version's logic (adapt now to be resilient; mitigate now to limit future impacts). The takeaway appears as "black mirror" in the first version and "smoke and mirrors" in the second; the workshop uses "black mirror." The second version's inquiry turns open questions into yes-or-no questions ("Has your understanding … changed?"); the polished text keeps the open forms.

Guiding questions in the workshop:

- When people come to understanding, people take action. (Part Three title slide)
- Which one of these could your class start this month? (three jobs at three scales: you, your family, your community)
- Which actions do more than one job? (the A, M, R sort)

4

PART FOUR

How Do We Work Together?

Relationships Matter

KEY LESSON

First version (verbatim):

> It's going to take all of us. We can do more together than any of us can alone, and that includes our relationship with our non-human relatives.

Polished:

It's going to take all of us. We can do more together than any of us can alone, and that includes our relationships with our non-human relatives.

KEY TAKEAWAY

First version (verbatim):

> We are not apart from nature, we are a part of nature. Go outside, pay attention, be a good relative, be a good ancestor. Our world is changing rapidly, but we have all that we need, and we are sovereign, and we decide what matters. We are not born into this world, we are born out of it.

Second version (verbatim):

> We are not born into this world, we are born out of it. We are not apart from nature, we are a part of nature. Our world is rapidly changing, but our sovereignty, and our knowledge will drive our communities to make healthier climate conscious decisions. Be a good relative, a good ancestor, a good human being.

Polished:

We are not born into this world; we are born out of it. We are not apart from nature; we are a part of nature. Our world is changing rapidly, but we have all that we need: we are sovereign, and we decide what matters. Go outside, pay attention, be a good relative, be a good ancestor.

DESCRIPTION

First version (verbatim):

> "We will always be better than I ever will be," the heart of this last model. The work and any action or thinking must go back to honoring our relationships and responsibilities, to each other and to the lands, waters, and all beings who live within them; the same way we think, by going outside to watch a cloud, we have to rely on our many senses. Experience shows us what is truly possible, Covid showed how fast the whole world can change when it has to, and the Israel and Iran war showed how quickly we can rethink our dependence on fossil fuels, so let us rely more upon what we have experienced moreso than what we "know."

Second version (verbatim):

> The work, the action, and the thinking must always be rooted in honoring our relationships and responsibilities to the land, the water, and all beings who live within them. Whether that be through reoccurring natural disasters, drought, war, famine, or airborne illness, let us rely more heavily on what we have experienced, rather than what we "know".

Polished:

"We will always be better than I ever will be" is the heart of this last part. The work, the action, and the thinking all go back to honoring our relationships and responsibilities: to each other, and to the lands, waters, and all Beings who live within them. The same way we learned by going outside to watch a cloud, we rely on our many senses. Experience shows us what is truly possible; we have watched the whole world change quickly when it had to. So let us rely more upon what we have experienced than on what we "know."

Notes: The first version names Covid and the Israel and Iran war as examples of fast change; the second generalizes to "natural disasters, drought, war, famine, or airborne illness." The polished text keeps the point without naming a war, so it does not date quickly or pull the room into current politics; restore either version's examples if you prefer. The source has no inquiry questions for Part Four; the workshop uses: What did you learn about the place of relationships in this work? What idea about being a good ancestor moved for you? What question will you keep working on? "Reoccurring" in the second version should read "recurring" if that version is used.

Guiding questions in the workshop:

- No single person, scientific report, or action plan has all the answers. (Part Four title slide)
- Which relative are you closest to, and who holds you up? (the web of relatives)

Language Guide

These choices apply across the workshop, lessons, and syllabi.

USE	INSTEAD OF	WHY
journey, homecoming, the journey home	road, route (for salmon)	The salmon are relatives on a journey, not traffic on a road
relatives	resources	Relatives, not resources
Tribes, Nations, Treaty Tribes, Tribal governments	stakeholders	Nations hold rights and govern; they are not one interest among many
reserved rights	granted rights	The Treaties reserved rights the Tribes already held (*U.S. v. Winans*, 1905)
Elder, Indigenous, Tribal, Tribes, Nations, Peoples, Treaty, First Foods	lowercase forms	Respect, and house style
Indigenous Knowledge	"Tribal Ecological Knowledge"	Matches current usage; TEK is "Traditional Ecological Knowledge" when used
the young ones, the next generation	juveniles (in spoken text)	Plain and relational; "juvenile" is fine in data tables
semicolons, commas, periods, or a new sentence	em dashes	House style
MM/DD/YYYY	other date formats	House style

Placeholders: anything only the authors or a Nation can supply stays in brackets, such as [PLACEHOLDER: names in the language of the host Nation, with permission].

Where Each Part Lives

PART	WORKSHOP SLIDES	GRADES 6 TO 12	CCIC 200	TCR 201
One: How does a cloud work?	3 to 6	Unit 1	Weeks 2 and 3	Module 1

PART	WORKSHOP SLIDES	GRADES 6 TO 12	CCIC 200	TCR 201
Two: How do you know what you know?	7 to 13	Unit 2	Weeks 4 and 6 (Week 5 is the bridge)	Module 2
Three: What can we do?	14 to 18	Unit 3	Weeks 7, 8, and 9	Module 3
Four: How do we work together?	19 to 21	Unit 4	Weeks 1, 4, and 10	Module 4

Sources and Credit

- Patrick Freeland and Cayla Jones, *Understanding a Climate of Change: Understanding an Ever Changing Climate*, PF Presentation tab, *Climate Change & The Tribes Curriculum 062326* (Google Doc), ATNI Climate.
- Steve Robinson, *Climate Change and the Tribes: Columbia River Edition* (draft), Tab 1 of the same document; case material, quotations, and Tribal examples.
- *Changing Currents Tribal Youth Water Summit Climate Change Curriculum* (2025 Youth Summit tab); the Elder-and-child elevator pitch.
- NESC 215, Northwest Indian College (catalog title: Climate Change in Native Lands; taught by Patrick Freeland as "Climate Change in Indian Country").